A Project Report

On

Computational Physics

BY

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Under the supervision of

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SUBMITTED IN PARTIAL FULLFILLMENT OF THE REQUIREMENTS OF

PHY F266 : Study Project



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE PILANI (RAJASTHAN)

HYDERABAD CAMPUS

September 2026

ACKNOWLEDGMENTS

I would like to express my sincere gratitude to Professor Sutapa Roy, my project supervisor, for her invaluable guidance, encouragement, and continuous support throughout the course of this work. Her insightful feedback, patience, and motivation have been instrumental in making the project better.

Finally, I am deeply grateful to my family for their constant encouragement and moral support, which kept me motivated throughout the project.



Birla Institute of Technology and Science-Pilani,

Hyderabad Campus

Certificate

This is to certify that the project report entitled "**Computational Physics**" submitted by Mr. Rudresh Asagodu (ID No. 2023B5A80772H) in partial fulfillment of the requirements of the course PHY F266, Laboratory Project Course, embodies the work done by them under my supervision and guidance.

Date: 29/09/26

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Abstract

Phase ordering Kinetics shows the evolution of an initially disordered system into an ordered state following a rapid quench. In this study project I studied the formation and growth of domains during this process, with emphasis on the role of domain walls and the characteristic length scale of the evolving system.

I studied about Ising ferromagnetic model, spinodal decomposition of binary alloys, that is, how system evolves for conserved number of atoms, I also studied about the Scaling hypothesis, allen cahn equation for conserved field, and growth of domain in non conserved field.

The order parameter and scaling hypothesis are used to understand the long time behavior of the domain growth of the system. This included the dynamics of the system with conserved and non conserved order parameters, described using the time-dependent Ginzburg–Landau and Cahn–Hilliard equations, respectively. For non conserved dynamics, domain growth is driven by curvature of domain walls, while for non conserved, it is driven by the diffusion of the order parameter through the system.

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Report on Computational Physics

Introduction

This project report presents a study on phase transitions, phase separation, domain growth, scaling behavior, domain walls, and conserved and non-conserved dynamics, with particular focus on the mathematical descriptions of these processes.

1) Phase Transition

A phase transition occurs when the system goes from one state to another, when a physical parameter, like temperature is varied. When a system is rapidly quenched from a high temperature disordered phase to a low temperature ordered one, the system does not become an ordered one instantaneously. Instead, different domains are formed which compete to select the equilibrium state. The process of gradual growth of domains with time is called domain coarsening.

2)Phase Diagram for Magnets

We consider the simplest, and the most familiar system: The ferromagnetic Ising model. Figure 1 shows a spontaneous magnetization as a function of temperature. The arrow indicates a temperature reduction to a final temperature Tf below critical temperature Tc from an initial temperature TI above Tc. From the figure it can be seen that at the final temperature there are two equilibrium phases, with magnetization +M0 and -M0. Now, right after the quench, this system is in an unstable disordered state compared to the initial temperature.

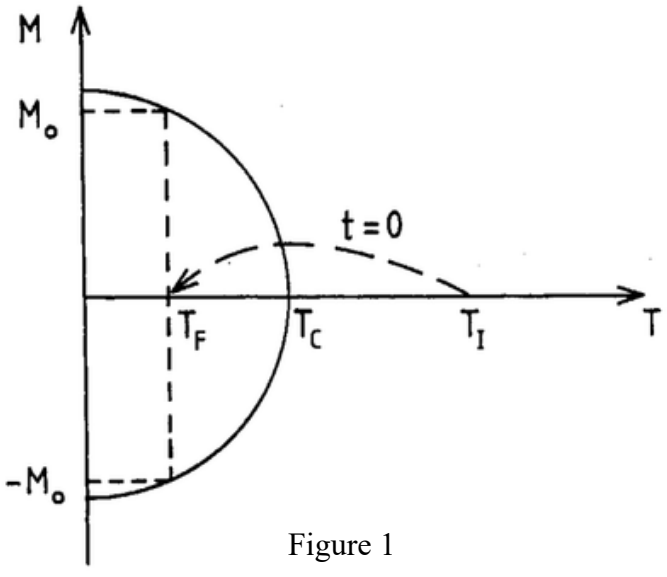


Figure 1

3) Binary liquids

When two liquids seperate into two phases, their domain also grows with time, just like the Ising model, but gravity can make the seperation happen too quickly, making it difficult to observe the normal domain-growth behaviour for a long time.

There is an extra complication that is not contained in the Ising model. If we represent the first liquid by up spin and the second liquid by down spin, then the equilibrium properties of this system can be modelled by the Ising model with Monte Carlo dynamics.. Now, flipping a spin in the Ising model corresponds to converting the first liquid to the second one which is not allowed or not possible. Therefore, the dynamics must conserve the amount of the first and second liquids. This causes slower growth than when the ordered parameter is non-conserved.

4) Phase seperation dynamics

Let $\phi(\mathbf{x}, t)$ be the order parameter field that describes the entire system.
The Landau free Energy is given as:

$$F[\phi] = \int d^d x \left[\frac{1}{2} |\nabla \phi|^2 + V(\phi) \right]$$

Where,$V(\phi)$ has a double well structure. The minima of $V(\phi)$ is taken to be at +1 and -1 and the minimum value is taken to be 0.

In the case where the order parameter is not conserved, an appropriate equation for the time evolution of the field is

$$\frac{\partial \phi}{\partial t} = -\frac{\delta F}{\delta \phi} = \nabla^2 \phi - V'(\phi)$$

When the order parameter is conserved, only diffusive transport of the ordfer parameter is allowd which causes the time evolution equation to become:

$$\frac{\partial \phi}{\partial t} = \nabla^2 \frac{\delta F}{\delta \phi} = -\nabla^2 \left[\nabla^2 \phi - V'(\phi) \right]$$

These equations are called the time-dependent Ginzburg Landau (TDGL) equation and the Cahn-Hilliard equation respectively.

In the above two equations there is no noise terms , this means that there is no thermal fluctuations, so we are effectively working at T=0. A schematic Renormalization Group(RG) flow diagram for T is given in figure 2, showing the three RG fixed points), Tc, infinity. Any point above Tc flows to infinity and any point below Tc flows to 0. So, the final temperature Tf is irrelevant in the scaling regime for quenches into the ordered phase

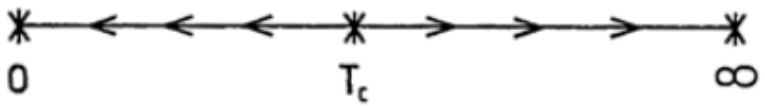


Figure 2

4.1) Scaling hypothesis

After a quench, the domains in the system grow with time. The scaling hypothesis assumes that, at sufficiently late times, the domain structure can be described by a single characteristic length scale L(t). This represents the typical size of the domains at time t.

Although the domains become larger as time increases, their overall statistical structure remains similar when distances are measured relative to L(t). Thus, the domain patterns at different times can be compared by rescaling their lengths using L(t).

Scaling has not yet been proved, except in some simple models, but the evidence are compelling
The domain structure can be described using the equal-time pair correlation function

$$C(r,t) = \langle \phi(\mathbf{x}+\mathbf{r},t)\phi(\mathbf{x},t)\rangle$$

and it's fourier transform, the equal - time structure factor,

$$S(\mathbf{k},t) = \langle \phi_{\mathbf{k}}(t)\phi_{-\mathbf{k}}(t)\rangle$$

Angular brackets indicate an average over the initial conditions.
The pair correlation function and the structure factor have the scaling forms:

$$C(r,t) = f\left(\frac{r}{L(t)}\right)$$

$$S(k,t) = L(t)^d g\left(kL(t)\right)$$

4.2) Domain Walls

A domain wall is the boundary seperating two domains with different values of the order paramater.
For a flat equilibrium domain wall, the wall profile satisfies:

$$\frac{d^2\phi}{dg^2} = V'(\phi)$$

Where g is the coordinate normal to the domain wall.
The boundary conditions are $\phi(-\infty) = -1, \quad \phi(+\infty) = +1$. The order parameter changes smoothly from -1 to +1. The excess energy is concentrated around the domain walls.
The energy per unit area, that is the surface area is given by:

$$\sigma = \int_{-\infty}^{\infty} dg \left(\frac{d\phi}{dg}\right)^2 = \int_{-1}^{1} d\phi \, [2V(\phi)]^{1/2}$$

Since the system tends to reduce its energy, domain growth occurs through a reduction in the total domain-wall area. The curvature of the domain walls provides the driving force for domain growth. This mechanism of growth is different for conserved and non-conserved order parameters.

5) Non conserved dynamics

The surface tension creates a force on the domain wall. This force is proportional to the local curvature of the wall.
For a spherical domain of radius R this force comes out as: F=2σ/R.
This force makes the domain wall move inward, reducing the domain size. For a non conserved order parameter, the wall velocity is proportional to the curvature:

η dR/dt=-2σ/R

Therefore, from here it can be seen that smaller domain shrink faster while larger domain change more slowly. In dimension d, the factor 2 becomes d-1.

Consider a spherical domain of radius R(t), and inside the domain φ = -1 and outside φ = +1.
For a spherical domain, we get: $0 = \frac{d-1}{R} + \frac{dR}{dt}$ This shows that the domain radius decreases with time.
Integrating the above equation we get t_collapse~R^2 (0) which says that larger domain takes longer to collapse.

For a curved domain wall , define $\hat{g}$ as the unit normal to the wall. This wall moves with velocity v in the normal direction.

The result for general curved surfaces that was defined by Allen and Cahn comes as:

$$v = -\nabla \cdot \hat{\mathbf{g}} = -K$$

K is d-1 times the mean curvature.
From here it can be said that the wall velocity is proportional to its curvature. The negative sign says that the wall shrinks. For a spherical domain K is (d-1)/R and the Allen Cahn equation simplifes to $0 = \frac{d-1}{R} + \frac{dR}{dt}$

•A second feature of Allen Cahn equation is that the surface tension σ does not appear.
Surface tension provides the driving force but the wall also experiences friction when it moves.
ηv=-σK - this is the equation of motion of the wall in the dissipative system and comparing with the Allen Cahn equation we see that η=σ.

The energy dissipation rate is $\frac{dE}{dt} = -\sigma v^2$
So, surface tension drives motion, while friction dissipates energy.

6) Conserved dynamics

Interfaces cannot move independently because the order parameter is conserved.At late times, the dominant growth occurs happens by the transport of the order parameter from interfaces of high curvature to regions of low curvature by diffusion.

Define $\phi = 1 + \tilde{\phi}$ where $\tilde{\phi}$ is a very small deviation about 1.
Now, at late times:

$$\frac{\partial \tilde{\phi}}{\partial t} = V''(1)\nabla^2 \tilde{\phi}$$

This is the diffusion equation with diffusion constant D = V''(1).
If the domain size is L, the diffusion field needs roughly $t_D \sim L^2$ to adjust across the domain. So for a large domain, diffusion takes a relatively long time. The interface velocity is

$v \sim \frac{1}{L^2}$

So during time t_D the interface barely moves. The diffusion field adjusts much faster than the domain walls move, i.e. the diffusion field is essentially always in equilibrium.
•In the bulk, the diffusion equation can eventually be replaced by Laplace's equation:

$$\nabla^2 \tilde{\phi} = 0$$

Define the chemical potential: **μ=δF/δϕ.** Conservation of ϕ gives the continuity equation **∂ϕ/∂t=-∇·j.**
The order-parameter flux is driven by the chemical-potential gradient: **j=-∇μ.**
Chemical potential with respect to ϕ is given by: $\mu = V'(\phi) - \nabla^2\phi$
μ also obeys laplace's equation **∇^2 μ=0**.
Near a curved interface, the chemical potential is: $\mu = V'(\phi) - \left(\frac{\partial\phi}{\partial g}\right)K - \frac{\partial^2\phi}{\partial g^2}$

Integrating across the interface gives the general boundary condition: μΔϕ=ΔV-σK. For equal depth minima ΔV=0,Δφ=2.
Therefore:

$$\mu = -\frac{\sigma K}{2}$$

This is Gibbs-Thompson boundary condition. The chemical potential at the interface depends on its curvature. Between interfaces, the chemical potential satisfies: $\nabla^2\mu = 0$
The interface moves due to the difference in chemical-potential gradients across it: $v\Delta\phi = -\left[\frac{\partial\mu}{\partial g}\right]$
Consider a spherical domain of radius R(t) in 3D. For a Sphere: K = 2/R.
From the Gibbs-Thompson condition: μ=-σK/2. Therefore, the chemical potential varies around the spherical domain and drives diffusion of material.

For a spherical domain, the chemical potential is: $\mu = \begin{cases} -\frac{\sigma}{R}, & r \le R \\ -\frac{\sigma}{r}, & r \ge R \end{cases}$
The interface velocity is: dR/dt=v=-σ/(2R^2)
Therefore: $t_{\text{collapse}} \sim R^3(0)$
Key difference is that in non conserved: $t_{\text{collapse}} \sim R^2$ while in conserved: $t_{\text{collapse}} \sim R^3$
Far from the interface, the chemical potential is: μ=V′(ϕ1)=V′(ϕ2). For a flat interface (K=0)the Gibbs–Thomson condition gives: μ=ΔV/Δϕ. Therefore V′(ϕ1)=V′(ϕ2)=ΔV/Δϕ.
This leads to the common tangent construction.

The points where the common tangent touches V(ϕ) determine:
φ1: order parameter of one phase.
φ2: order parameter of the other phase
For a spherical domain, this gives the general domain-growth equation: $\frac{dR}{dt} = -\frac{2\sigma}{\Delta\phi\, R^2}$

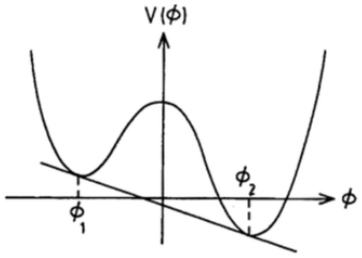


Figure 3

7) Conclusion

This study helped in understanding the evolution of a system from a disordered state to an ordered state following a quench. The formation and growth of domains were studied using the Ising model and phase separation in binary liquids. The role of the order parameter and the characteristic length scale L(t) was examined through the scaling hypothesis, which provides a way to describe the domain structure at late times.

The study also provided an understanding of domain walls and how their curvature and surface tension drive domain growth. For non conserved dynamics, domain walls move due to curvature, leading to the reduction of domain-wall area. For conserved dynamics, the order parameter cannot change locally and domain growth occurs through diffusion driven by differences in chemical potential.

Overall, the study showed how the mathematical descriptions of conserved and non-conserved dynamics lead to different mechanisms of domain growth. The concepts of scaling, domain-wall motion, diffusion, and chemical potential provide a framework for understanding the long-time evolution of phase separating systems.

8) References

Bray, A. J., "Theory of phase-ordering kinetics". Advances in Physics, 43(3), 357–459 (1994).